SUMMER PROJECT RESEARCH REPORT

Multimodal Understanding of Comics and Visual Narratives: A Literature Review and Research Framework

A Literature-Based Study of Multimodal Understanding, Visual Narrative Reasoning, and Future Research Directions

Submitted by

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Declaration

I hereby declare that this report presents an original literature-based study of multimodal understanding of comics and visual narratives. The work is based primarily on the study, analysis, and synthesis of relevant research literature.

The study focuses on problem formulation, analysis of existing approaches, identification of research gaps, and development of a conceptual research framework. No experimental implementation, model training, testing, or quantitative evaluation is reported in this work.

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Finally, I would like to thank everyone who contributed directly or indirectly to the completion of this work.

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Abstract

Comics represent a distinctive form of visual narrative in which meaning is conveyed through the interaction of images, textual elements, spatial organization, and sequential structure. Unlike conventional image-understanding tasks, computational understanding of comics requires systems to reason about relationships between multiple panels, characters, textual elements, and events. In particular, important narrative information may remain implicit in the transitions between panels, requiring the inference of events that are not explicitly depicted.

This work presents a literature-based study of multimodal understanding of comics and visual narratives. Twelve research papers covering comic understanding, panel extraction, reading-order analysis, optical character recognition, character and dialogue understanding, multimodal reasoning, emotion recognition, object detection, and narrative generation were studied and analyzed. The literature was examined thematically to identify the major computational components involved in understanding visual narratives and to determine the limitations of existing approaches.

Based on the literature analysis, a conceptual research framework is formulated that progresses from page and layout understanding through panel detection, visual and textual information extraction, character–text association, multimodal fusion, sequential reasoning, and narrative reconstruction. The framework is proposed as a research direction for integrating the complementary capabilities identified across the reviewed studies.

The study does not report experimental implementation or quantitative evaluation. Instead, its primary contribution is the systematic synthesis of existing research, identification of major research gaps, and formulation of a conceptual framework that can guide future experimental investigation in multimodal comic and visual-narrative understanding.

Keywords: Comics Understanding, Multimodal Learning, Vision-Language Models, Visual Narratives, Comic Analysis, Narrative Reasoning, OCR, Sequential Reasoning

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1 Introduction

1.1 Background

Visual narratives represent a form of communication in which meaning is conveyed through the interaction of visual elements, textual information, spatial organization, and temporal or sequential structure. Unlike conventional images, where the primary objective of an artificial intelligence system may be to recognize objects or generate a description of a scene, visual narratives require the interpretation of relationships between multiple visual units and the reconstruction of the underlying events or story.
Comics provide a particularly interesting example of such visual narratives. A comic combines illustrations, characters, objects, speech balloons, narration boxes, visual symbols, and panel layouts to communicate a story. Importantly, the complete meaning of a comic is often not explicitly represented within any individual panel. Instead, readers infer connections between panels and construct a coherent narrative from the information presented across the sequence.
This characteristic makes comics a challenging domain for computational understanding. Recent research describes comics as a medium that combines rich visual and textual narratives while introducing additional challenges arising from stylistic variation, complex page layouts, changing reading orders, and non-linear storytelling [1].
The difficulty becomes particularly apparent when considering the role of the space between consecutive panels, commonly referred to as the *gutter*. Actions, changes in location, changes in time, and other events may not be explicitly depicted. Instead, the reader is expected to infer these missing events from the surrounding panels. Iyyer et al. describe this process as *closure*, through which readers connect individual panels and infer unseen actions to form a coherent storyline [2].
Therefore, computational comic understanding cannot be reduced to recognizing the objects appearing in individual images. A successful system needs to combine information from multiple modalities and reason over the relationships between them.

1.2 Comics as Multimodal Visual Narratives

A comic page can be viewed as a multimodal representation consisting of several interacting information sources.
These include:

- **Visual information** — characters, objects, environments, actions, and visual symbols;
- **Textual information** — dialogue, narration, captions, and other textual elements;
- **Spatial information** — panel boundaries, locations of speech balloons, and relative positions of elements;

- **Sequential information** — the ordering of panels through which the story progresses;
- **Contextual information** — relationships between characters, objects, and events across panels.

The interaction between these components is fundamental to understanding comics.
For example, a dialogue statement may be ambiguous without knowing which character is speaking. Similarly, an image may contain an action whose significance becomes clear only after examining the preceding panel. Consequently, the visual and textual modalities cannot always be interpreted independently.
This observation is strongly supported by the COMICS study. Analysis of the dataset showed that more than 90% of panels relied on both text and image to convey information. The authors therefore argued that models attempting to understand comic narratives must connect multimodal information and perform inference across panel transitions [2].
The problem can therefore be represented conceptually as

$$\text{Visual Information} + \text{Textual Information} + \text{Spatial Structure} + \text{Sequential Context} \longrightarrow \text{Narrative Understanding}. \tag{1}$$

This perspective forms the foundation for the research direction explored in this study.

1.3 Computational Understanding of Comics

The application of artificial intelligence to comics has traditionally been divided into a number of individual computational tasks.
These include:

1. panel detection and segmentation;
2. character detection;
3. object detection;
4. text and speech-balloon detection;
5. optical character recognition (OCR);
6. character–text association;
7. reading-order determination;
8. emotion recognition;
9. multimodal reasoning;

10. narrative inference and generation.

Recent survey research demonstrates that comics understanding has developed into a broad vision-language research area containing tasks ranging from basic image processing and object recognition to complex multimodal narrative reasoning [1].
The survey introduces the *Layers of Comics Understanding* (LoCU) framework to organize the different tasks involved in comics intelligence. This demonstrates that understanding a comic is not a single computer-vision problem but rather a collection of interconnected computational problems [1].
A simplified representation of the overall process is shown below.

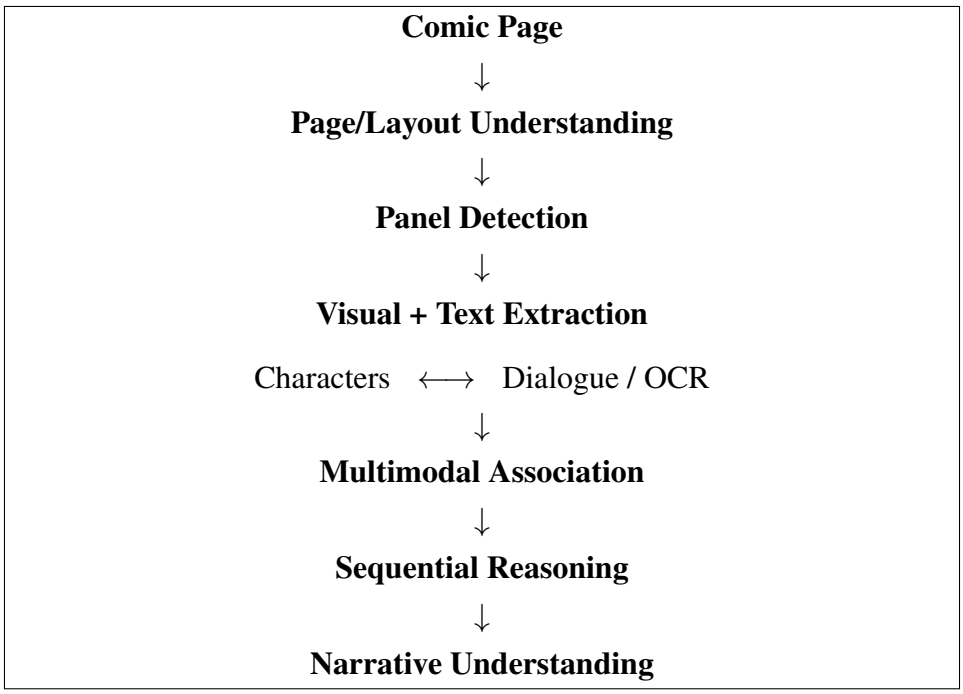


Figure 1: Conceptual decomposition of computational comic understanding.

Each stage provides information required by subsequent stages.

1.4 Challenges in Comic Understanding

Several characteristics distinguish comics from conventional computer-vision datasets.

1.4.1 Artistic and Stylistic Variation

Comics can vary substantially in drawing style, color, line structure, character representation, and visual abstraction. The same semantic object can therefore have very different visual appearances across different comics.
The recent literature identifies stylistic variation as one of the fundamental challenges in applying machine-learning techniques to comics [1].

1.4.2 Complex Page Layouts

Comic pages do not necessarily follow a fixed grid. Panels can have different sizes and shapes, and elements may overlap, bleed beyond panel boundaries, or be arranged in unconventional ways. Consequently, simply applying standard rectangular object detection to a comic page may not be sufficient for accurate structural understanding.
The segmentation literature identifies grids, staggered layouts, bleeds, and insets as examples of layout complexity encountered in comic pages [1].

1.4.3 Reading Order

Unlike ordinary documents, the reading order of panels cannot always be assumed to be a simple left-to-right and top-to-bottom sequence.
This introduces a structural problem:

$$\text{Page Image} \rightarrow \text{Panels} \rightarrow \text{Correct Sequence}. \tag{2}$$

Incorrect reading order can subsequently lead to incorrect narrative interpretation.

1.4.4 OCR and Textual Variation

Comic text introduces additional challenges for OCR systems because of stylized fonts, unusual typography, capitalization, dialects, and the placement of text within speech balloons or narration boxes.
The COMICS dataset construction study reported that several conventional OCR systems struggled with comic text, particularly because of font variation and domain mismatch [2].
Therefore,

$$\text{OCR Output} \neq \text{Complete Textual Understanding}. \tag{3}$$

The system must also determine where the text belongs and which character or narrative element it is associated with.

1.4.5 Character Consistency and Coreference

Characters can appear repeatedly throughout a comic, sometimes from different viewpoints, with different poses, expressions, and occlusions.
A computational system must therefore determine whether two visual appearances correspond to the same character.
This becomes particularly important for narrative reasoning because events involving the same character need to be connected across multiple panels.
Character consistency and coreference have consequently emerged as important components of

comics understanding research [1].

1.4.6 Inter-Panel Reasoning

Perhaps the most important challenge is that the information necessary to understand a story may be distributed across multiple panels.
For example,

$$P_1 \rightarrow P_2 \rightarrow P_3 \tag{4}$$

may represent the following conceptual sequence:

$$\text{Action}_1 \rightarrow \text{Hidden Event} \rightarrow \text{Consequence}_3. \tag{5}$$

The hidden event is not necessarily visible in any panel. The system must infer it from contextual information.
This is the central motivation behind the closure-based tasks introduced in the COMICS dataset, where models are required to use preceding panels to reason about subsequent narrative and character-centric information [2].

1.5 Motivation for the Present Study

The increasing availability of vision-language models has created new opportunities for computational understanding of complex visual narratives.
However, the literature indicates that comics remain a challenging domain compared with conventional natural-image tasks. Existing studies address many individual components, including detection, segmentation, OCR, character linking, and multimodal reasoning, but these components do not necessarily constitute a unified narrative-understanding system [1].
The recent survey literature also emphasizes the diversity of tasks and datasets in comics understanding and the need for further research into more complex multimodal tasks [1].
This motivated the literature-based study presented in this work.
The objective was therefore not simply to investigate how an artificial intelligence model could recognize objects in comic images, but to understand the broader research problem:

> *How can visual, textual, spatial, and sequential information be integrated to enable machines to understand the narrative conveyed by comics?*

1.6 Problem Statement

Based on the literature examined in this study, the research problem can be formulated as follows:

> **To investigate computational approaches for multimodal understanding of comics by studying how page layout, panel structure, visual elements, textual information,**

character identity, and sequential relationships can be integrated to support narrative understanding and reconstruction.

The problem can be decomposed into the following sub-problems:

$$
\boxed{\begin{array}{l}
\text{Panel / Region Detection} \\
\downarrow \\
\text{Reading-Order Determination} \\
\downarrow \\
\text{Visual Element Understanding} \\
\downarrow \\
\text{Text Extraction and OCR} \\
\downarrow \\
\text{Character–Text Association} \\
\downarrow \\
\text{Multimodal Representation} \\
\downarrow \\
\text{Sequential Reasoning} \\
\downarrow \\
\text{Narrative Reconstruction}
\end{array}} \tag{6}
$$

This decomposition will subsequently be used to construct the conceptual research framework discussed in Section 4.

2 Fundamentals of Multimodal Comic Understanding

2.1 Introduction

Comics are a form of visual narrative in which images, text, spatial organization, and sequential relationships work together to communicate meaning. Unlike conventional natural images, a comic page is not simply a collection of independent visual objects. Its interpretation depends on the relationships among panels, characters, textual elements, and events across a sequence.
Recent research describes comics as a particularly challenging domain for vision and language systems because they combine rich visual and textual information with stylistic variation, complex reading orders, and sequential narrative structures [1]. These characteristics make comic understanding an inherently multimodal problem.
This section introduces the fundamental concepts required to understand the research discussed in the subsequent sections. It covers comics as a visual language, page and panel structure, reading

order, computer vision, object and character detection, image segmentation, optical character recognition, multimodal learning, vision–language models, sequential reasoning, and narrative representation.

2.2 Comics as Visual Language

Comics communicate stories through the coordinated use of visual and textual elements. A typical comic page may contain multiple panels, characters, objects, backgrounds, speech bubbles, captions, and other graphical elements. The meaning of an individual element is often influenced by its surrounding elements and by its position within the overall sequence.
The recent survey on comics understanding describes comics as a medium that combines visual and textual narratives and emphasizes that their unique structure introduces challenges beyond conventional image understanding [1].
A useful conceptual representation of a comic page is

$$P = \{p_1, p_2, \ldots, p_n\}, \tag{7}$$

where each p_i represents a panel within the page.
Each panel may contain several forms of information:

$$p_i = (V_i, T_i, E_i, R_i), \tag{8}$$

where V_i represents visual information, T_i represents textual information, E_i represents entities such as characters and objects, and R_i represents relationships among these elements.
This representation emphasizes that a comic panel cannot always be understood by considering only one modality.

2.3 Comic Page Layout and Panel Structure

2.3.1 Panels as Fundamental Structural Units

A comic page is commonly divided into panels that represent individual moments, scenes, or stages of a narrative. Panels provide the structural organization through which a reader observes the progression of events.
Automatic panel extraction is therefore an important foundational task in computational comic understanding. Pang et al. specifically investigated the problem of extracting panels from manga pages containing irregular, unclosed, borderless, or joined panel structures [3].
The problem can be represented conceptually as identifying a set of regions

$$\mathcal{P} = \{R_1, R_2, \ldots, R_n\}, \tag{9}$$

where each region R_i corresponds to a detected panel.

Correct panel identification is important because errors at this stage can propagate to later tasks such as reading-order prediction, OCR, character analysis, and narrative reasoning.

2.3.2 Variation in Panel Layout

Comic layouts are not necessarily uniform. Panels may differ in size, aspect ratio, shape, and spatial arrangement. The global analysis of comics presented by Cohn et al. demonstrates substantial variation in panel frequency and size across comics from different regions and contexts [4]. Therefore, a computational system should not assume that every comic page follows the same structural layout.

The difficulty is further illustrated by research on digital comic interfaces. Different presentation formats may preserve or modify the original relationships between panels, which can influence how readers navigate and interpret the narrative [5].

2.4 Reading Order and Sequential Organization

Reading order refers to the sequence in which panels and their contents are interpreted by a reader. A simple representation of a page with n panels is

$$\pi = (p_{\pi_1}, p_{\pi_2}, \ldots, p_{\pi_n}), \tag{10}$$

where π represents the ordering of panels.

Determining π is not always equivalent to sorting panels according to their spatial coordinates. Cohn's empirical study of comic-page navigation demonstrated that readers use structural principles when navigating complex layouts [6].

The problem becomes more complex when panels overlap, differ considerably in size, or are arranged according to unconventional visual compositions.

Reading conventions can also differ across comic traditions. The literature therefore suggests that reading order should be considered a structural and contextual problem rather than a fixed geometric rule [6, 4].

Reading order is particularly important for computational narrative understanding because an incorrect sequence can change the interpretation of events.

2.5 Computer Vision for Comic Analysis

Computer vision provides the basic mechanisms required to extract information from comic images. In conventional natural-image analysis, computer-vision systems may identify objects, regions, faces, scenes, or other visual features. Similar techniques can be applied to comics, but comic imagery introduces additional challenges.

The survey on cartoon image processing notes that cartoon images differ from natural photographs through properties such as clear lines, smooth colour regions, and simplified or stylized visual

representations [7].
The comics-understanding literature further identifies tasks such as object detection, semantic segmentation, character recurrence, pose estimation, and other forms of visual analysis as relevant components of computational comic understanding [1].
Thus, computer vision serves as a foundation for extracting the visual evidence required by later multimodal reasoning stages.

2.6 Object and Character Detection

Object detection aims to identify objects within an image and determine their locations.
A detected object can be represented as

$$o_i = (c_i, b_i, s_i), \tag{11}$$

where c_i denotes the predicted class, b_i denotes the bounding box, and s_i represents the associated confidence score.
In comic understanding, the detected entities may include characters, animals, vehicles, objects, or other visually important elements.
Character detection is especially important because characters frequently reappear throughout a sequence. Consequently, recognizing a character in one panel is not necessarily sufficient; the system may also need to determine whether a visually similar character appearing in another panel represents the same entity.
The literature on comics understanding includes character detection and recurrence among the important tasks required for higher-level comic interpretation [1]. Recent comic-to-prose research also incorporates character detection, clustering, and grounding as part of a larger multimodal pipeline [8].
Generic object-detection models may provide useful foundations for such tasks. However, the stylized nature of comics means that performance obtained on natural-image datasets cannot automatically be assumed to transfer to comic images [7, 9].

2.7 Image Segmentation

Object detection generally represents objects using bounding boxes, whereas image segmentation attempts to identify the precise pixels or regions belonging to particular objects or structures.
For a comic image I, a segmentation procedure can be viewed conceptually as assigning a label to each pixel:

$$S(x, y) \in \mathcal{C}, \tag{12}$$

where $\mathcal{C}$ represents the set of possible region or object classes.
Segmentation can be useful for identifying characters, objects, panels, or other visual regions.

The comics-understanding survey identifies instance segmentation and related visual tasks among the broader set of computational problems encountered in comic analysis [1].
For panel analysis, segmentation is particularly useful when the boundary of a panel cannot be represented reliably through simple rectangular detection.

2.8 Optical Character Recognition

Optical Character Recognition (OCR) refers to the extraction of textual information from images.
In comics, OCR may be required to identify:

- speech-bubble dialogue;
- captions;
- narration;
- signs;
- labels; and
- other textual elements.

A simplified representation of OCR can be written as

$$I_{\text{text}} \rightarrow T, \tag{13}$$

where I_{text} represents a visual text region and T represents the recognized textual sequence.
Comic OCR can be considerably more difficult than ordinary document OCR because text may appear in stylized fonts, unusual orientations, irregular locations, or visually complex backgrounds.
The COMICS research demonstrated that OCR quality can directly influence higher-level comic reasoning. Its error analysis reported cases in which poor OCR quality or low image resolution contributed to errors even in human evaluation [2].
Recent comic-to-prose research also places OCR within a larger pipeline, extracting dialogue in reading order before using the resulting information for narrative generation [8].
Thus, OCR should be considered an information-extraction component rather than the complete text-understanding stage.

2.9 Multimodal Learning

Multimodal learning concerns the integration of information from multiple modalities.
In the context of comics, the two most important modalities are generally visual information and textual information.
Let

$$V = F_v(I) \tag{14}$$

represent a visual representation extracted from an image I, and

$$T = F_t(X) \tag{15}$$

represent a textual representation extracted from textual content X.
A multimodal representation can then be expressed conceptually as

$$Z = F(V, T), \tag{16}$$

where F denotes a multimodal fusion or reasoning mechanism.
The objective is not merely to place two representations together, but to allow information from one modality to help interpret the other.
For example, visual information may identify the character present in a scene, while dialogue may provide information about that character's intention or response.
The comics-understanding survey describes the broader transition from independent unimodal representations toward alignment and integration of multiple modalities in modern vision–language systems [1].
Similarly, the COMICS study demonstrated the value of jointly using image and text information for comic narrative tasks [2].

2.10 Vision–Language Models

Vision–Language Models (VLMs) are models designed to process or relate visual and textual information.
Modern vision–language systems have been applied to tasks such as image captioning, visual question answering, multimodal retrieval, reasoning, and generation. The recent comics-understanding survey identifies comics as a particularly challenging application domain for these models because comic understanding combines visual and textual information with sequential reasoning [1].
For a comic panel, a VLM can conceptually be viewed as receiving

$$Z_i = f(I_i, T_i), \tag{17}$$

where I_i represents the panel image and T_i represents associated textual information.
The model may then produce an output Y_i such as an answer, description, classification, or structured representation.
However, a complete comic often requires reasoning over several panels:

$$Y = f(I_1, T_1, \ldots, I_n, T_n). \tag{18}$$

This makes comic understanding more challenging than ordinary single-image vision–language tasks.
Recent benchmark research further demonstrates that large multimodal models can struggle with comic-specific tasks requiring sequence and contextual reasoning [10].

2.11 Sequential Reasoning

Sequential reasoning refers to the ability to interpret information across an ordered sequence rather than treating each element independently.
For a sequence of panels

$$P = (p_1, p_2, \ldots, p_n), \tag{19}$$

the interpretation of a panel may depend on information from earlier panels:

$$H_i = f(p_1, p_2, \ldots, p_{i-1}), \tag{20}$$

where H_i represents the contextual history available when interpreting panel p_i.
The resulting interpretation can therefore be represented as

$$R_i = f(p_i, H_i). \tag{21}$$

This is particularly important in comics because not every event is explicitly shown. Information can be distributed across panels, requiring the reader to infer relationships between them.
The COMICS study directly investigated this problem through tasks involving textual and visual closure and character coherence. Its findings showed that context from preceding panels can help performance, while strong but inappropriate image–text associations can still mislead models [2].
The ComicsPAP benchmark similarly evaluates sequence filling, text closure, visual closure, character coherence, and caption relevance, emphasizing the importance of sequential reasoning in comic understanding [10].

2.12 Narrative Representation

Narrative representation refers to the structured representation of events, entities, relationships, and contextual information that together describe a story.
A simple conceptual representation of a comic narrative can be expressed as

$$N = \{E, C, O, R, T\}, \tag{22}$$

where:

- E represents events or actions;
- C represents characters;
- O represents relevant objects;
- R represents relationships among entities and events; and
- T represents temporal or sequential information.

Narrative representation is a higher-level objective because it depends on the correct interpretation of several lower-level components.
For example, identifying a character and extracting a dialogue sentence are individual recognition tasks. Determining that the identified character spoke the sentence, understanding what action occurred before the dialogue, and relating the event to the following panel constitute narrative reasoning.
Recent research on generating literary narratives from comics illustrates this transition from structured comic information to coherent prose. The proposed pipeline combines panel, character, text, and speech-related information before generating a narrative description [8].

2.13 The Gutter and Implicit Information

One distinctive characteristic of comics is the space between panels, commonly referred to as the gutter.
The gutter may contain no explicit visual representation of an event, yet the reader may infer a transition between the preceding and succeeding panels.
For example, two panels may depict a character before and after an action without explicitly showing the action itself. Understanding the sequence therefore requires inference rather than simple visual recognition.
This concept is central to the COMICS study, which investigates inference between panels through narrative closure tasks [2].
The problem is related to the broader concept of underspecification in multimodal systems, where different modalities may leave aspects of an event unstated or ambiguous [11].
Consequently, a complete comic-understanding system must distinguish between information that is directly observed and information that is inferred from context.

2.14 Relationship Between the Fundamental Components

The concepts introduced in this section can be viewed as a hierarchy.

$$\text{Comic Page} \rightarrow \text{Panel Structure} \rightarrow \text{Reading Order} \rightarrow \text{Visual/Text Understanding} \rightarrow \text{Entity Association} \rightarrow \text{Multimodal Fusion} \rightarrow \text{Sequential Reasoning} \rightarrow \text{Narrative Understanding}. \tag{23}$$

At the lower levels, the objective is to identify and organize information. At higher levels, the objective shifts toward integrating this information and reasoning about relationships across the narrative.

This hierarchy provides the conceptual foundation for the research framework developed in Section 4.

3 Literature Review

3.1 Introduction

Computational understanding of comics represents a challenging intersection of computer vision, natural language processing, and multimodal reasoning. Unlike conventional natural images, comics communicate information through a combination of illustrations, characters, objects, speech balloons, narration boxes, visual symbols, panel layouts, and sequential relationships. Consequently, understanding a comic page requires more than recognizing the objects appearing in an individual image. A system must also determine how visual and textual elements interact and how information is carried across successive panels.

The literature considered in this study reflects the development of the field from relatively specialized image-processing tasks toward more integrated multimodal and narrative understanding systems. The studied works include methods for panel extraction, page navigation, character and object understanding, optical character recognition (OCR), emotion recognition, multimodal reasoning, comic-specific benchmarking, and narrative generation.

A useful way of organizing these contributions is according to the different stages involved in computational comic understanding. At the lower levels, a system must identify the structural units of a comic page, particularly panels and their spatial organization. At subsequent levels, it must identify characters, objects, text, and other visual elements. These elements then need to be associated with one another, after which multimodal and sequential reasoning can be performed. At the highest level, the system may attempt to reconstruct or generate a coherent narrative from the available visual and textual evidence.

The following sections review the literature according to these functional themes rather than presenting the papers chronologically.

3.2 Computational Understanding of Comics

Comics are a distinctive form of sequential visual narrative in which visual and textual information jointly contribute to meaning. The survey by Vivoli et al. [1] provides a comprehensive view of the field and emphasizes that comics understanding encompasses a large variety of tasks, ranging from basic image classification and object detection to visual reasoning and narrative generation.
The survey introduces the *Layers of Comics Understanding* (LoCU) framework as a taxonomy for organizing vision-language tasks related to comics. The framework includes tasks such as page segmentation, object detection, character re-identification, character–balloon association, dialog transcription, instance segmentation, visual question answering, visual reasoning, image-to-text generation, and scene-graph generation. This taxonomy demonstrates that comics understanding is not a single computer vision problem but a collection of interdependent tasks [1].
The complexity of the problem is partly caused by the distinctive properties of comics. Comic artwork can vary substantially in drawing style, panel composition, visual abstraction, and reading direction. Moreover, the order in which panels should be interpreted is not always recoverable from simple spatial coordinates. The survey therefore identifies comics as a particularly demanding domain for vision-language models.
The work of Iyyer et al. [2] provides an important foundation for understanding this complexity. The authors argue that comic narratives contain explicit information as well as deliberately omitted events. Readers must infer events occurring between adjacent panels, a process referred to as *closure*. Their COMICS dataset contains more than 1.2 million panels obtained from almost 4,000 Golden Age American comic books published between 1938 and 1954. Their experiments demonstrate that neither textual information nor visual information alone is sufficient to fully understand a comic narrative.
This observation is central to the literature reviewed in this study: comic understanding requires integration of multiple information sources rather than isolated processing of individual modalities.

3.3 Comic Page and Panel Segmentation

Before semantic interpretation can take place, a comic page must be decomposed into meaningful visual units. The panel is one of the most important such units because it typically represents a single temporal or narrative state within the larger story.
Pang et al. [3] investigated automatic panel extraction from manga pages. Their work addresses the difficulty caused by irregular, borderless, and visually complex panel layouts. Rather than assuming that every panel is a simple rectangular region, their method first identifies a panel block and repairs open panel boundaries. It then recursively divides the page using adaptive splitting lines before recovering the shapes of individual panels.
The method consists broadly of three stages: panel-block generation, recursive panel-block splitting, and panel-shape extraction. This approach demonstrates that panel segmentation cannot always be reduced to conventional rectangular object detection. Manga pages may contain irregular

boundaries, connected visual regions, and stylistic elements that interfere with straightforward segmentation [3].
Panel segmentation is therefore not merely a preprocessing operation. Errors at this stage can propagate to later components. If two panels are incorrectly merged, or if a single panel is incorrectly divided, subsequent OCR, character recognition, reading-order estimation, and narrative reasoning may all be affected.
The broader survey by Vivoli et al. [1] places page stream segmentation and related structural tasks within the lower layers of comics understanding. This reinforces the importance of correctly identifying the page structure before attempting higher-level semantic interpretation.

3.4 Reading-Order Determination

Once panels have been extracted, the next structural problem is determining the order in which they should be interpreted. Although a simple left-to-right and top-to-bottom ordering may work for some pages, comic layouts can contain irregular arrangements, overlapping panels, nested structures, and culturally dependent reading conventions.
Cohn [6] investigated how readers navigate comic page layouts and demonstrated that reading order cannot be completely explained by a simple spatial sorting rule. Participants were asked to number panels according to their perceived reading order. The results showed strong preferences for particular starting locations and different strategies depending on page structure and reader experience.
The study proposes that comic-page navigation is influenced by hierarchical and structural constraints rather than being determined solely by Cartesian coordinates. In particular, readers construct relationships between panels and larger page-level constituents when determining how the page should be traversed.
This finding is important for computational systems because a model that simply sorts panels according to their x and y coordinates may produce incorrect narrative sequences. Reading order can depend on panel geometry, visual grouping, cultural convention, and the overall composition of the page.
The global analysis of comic panels by Cohn et al. [4] provides additional evidence for substantial structural diversity. Their TINTIN corpus contains more than 76,000 panels from 1,030 comics representing 144 countries and territories. Their analysis shows variation in panel frequency and size across different regions, styles, languages, and publication periods.
Consequently, a robust computational framework should not assume that a single fixed page-layout rule is universally applicable.

3.5 Character and Object Understanding

After page decomposition, semantic interpretation requires identifying the entities represented within each panel. Characters are particularly important because they provide continuity across

successive panels and often serve as the participants in the narrative.
Object detection provides a general mechanism for locating meaningful visual entities within an image. Gupta et al. [9] investigated an optimized YOLO-NAS-based object detection framework, demonstrating the broader relevance of modern object detection architectures to visual recognition tasks.
However, generic object detection should not be regarded as a complete solution to comic understanding. Comic artwork differs from natural photographic imagery in terms of line structure, color distribution, stylization, deformation, and abstraction. Zhao et al. [7] review these differences in their survey of cartoon image processing, highlighting the need for methods capable of handling the particular properties of stylized visual content.
Character understanding also requires maintaining identity across multiple panels. A character may appear at different scales, orientations, poses, or with substantial changes in surrounding context. Therefore, character detection within a single panel and character re-identification across panels represent related but distinct problems.
The comics-understanding taxonomy of Vivoli et al. [1] explicitly includes character re-identification as well as object detection, indicating that identity continuity is an important component of higher-level comics understanding.

3.6 Optical Character Recognition and Text Understanding

Text is an essential component of comic narratives. Dialogue is frequently represented through speech balloons, while narration boxes, captions, sound effects, and other textual elements may provide additional information.
OCR in comics is considerably different from conventional document OCR. Text may be curved, rotated, stylized, partially occluded, placed over complex backgrounds, or represented using fonts that differ substantially from standard document datasets.
Iyyer et al. [2] incorporated automatically generated textbox transcriptions into their COMICS dataset. Their work also illustrates why text alone is insufficient: dialogue may refer to characters, objects, and actions that are visually represented but not explicitly described by the text.
The comics-understanding survey identifies dialog transcription as a specific task within the comics-understanding pipeline, alongside character–balloon association and character grounding [1]. This distinction is important because extracting text is only the first step. A complete system must also determine which character produced a particular dialogue and how that dialogue relates to the surrounding visual context.
Thus, OCR should be regarded as a component of multimodal understanding rather than as an isolated end goal.

3.7 Character–Text Association

A further challenge arises from the need to connect textual elements with the correct visual entities. In a typical comic panel, several characters may be present simultaneously, while multiple speech balloons may occur in different locations.
The problem therefore involves establishing a mapping

$$\text{Character} \longleftrightarrow \text{Speech Balloon} \tag{24}$$

before dialogue can be reliably interpreted as part of a narrative.
The comics-understanding taxonomy of Vivoli et al. [1] identifies character–balloon association, also referred to as speaker identification, as a distinct task. This demonstrates that speech recognition and character identification are not independent problems.
The more recent "From Panels to Prose" framework by Sachdeva et al. [8] places character grounding and dialogue-speaker association within a unified comic-understanding pipeline. Their approach jointly considers characters, text regions, panels, and speech-balloon relationships before attempting narrative generation.
This direction is particularly relevant to the conceptual framework developed in this research because it shows how individual recognition tasks can be combined into a structured representation of the comic page.

3.8 Emotion and Contextual Understanding

Beyond recognizing objects and text, comic understanding requires interpreting semantic and affective information. Characters' emotions, actions, and interactions may be communicated through a combination of facial expressions, body posture, visual symbols, and dialogue.
Dutta et al. [12] proposed EmoComicNet, a multitask multimodal framework for emotion recognition in comics. The model combines visual and textual information and uses relevance-aware fusion rather than simply concatenating the two modalities.
The significance of this work extends beyond emotion classification. It illustrates a general principle of comic understanding: different modalities can contain complementary evidence. A facial expression may provide information that is absent from dialogue, while dialogue may clarify an emotional state that is visually ambiguous.
This principle is consistent with the findings of Iyyer et al. [2], where the combination of image and text is necessary for understanding narrative relationships.
Consequently, semantic interpretation of a comic page should ideally consider visual appearance, textual content, character identity, spatial relationships, and contextual information jointly.

3.9 Multimodal Fusion

Multimodal fusion is a central component of computational comic understanding because the meaning of a panel is distributed across multiple representational channels.
A simplified representation of a comic panel can be written as

$$P_i = \{V_i, T_i, C_i, S_i\}, \tag{25}$$

where V_i represents visual information, T_i represents textual information, C_i represents character and object information, and S_i represents structural or spatial information.
A multimodal system must combine these components to produce a representation

$$H_i = f(V_i, T_i, C_i, S_i), \tag{26}$$

where H_i denotes the integrated representation of panel i.
The precise form of $f(\cdot)$ depends on the architecture. Earlier systems may use feature concatenation or task-specific fusion mechanisms, whereas contemporary vision-language models can jointly process image and text tokens.
EmoComicNet provides an example of relevance-aware multimodal fusion for comic emotion recognition [12]. The work demonstrates that the contribution of a modality may vary depending on the available evidence.
Iyyer et al. [2] provide an even broader narrative motivation for multimodal fusion. Their analysis indicates that neither text nor image independently contains enough information to completely represent the comic storyline.
Therefore, multimodal fusion should not simply be viewed as a method for improving classification accuracy. It is fundamental to representing the way comics themselves communicate information.

3.10 Sequential Reasoning and Narrative Closure

One of the defining properties of comics is that individual panels usually do not constitute complete narratives by themselves. Instead, the reader constructs continuity across panels.
Iyyer et al. [2] formalized this challenge through the concept of *closure*. Their work introduced three cloze-style tasks designed to test narrative and character-centric reasoning from preceding panels. The results showed that deep neural models underperformed human baselines, highlighting the difficulty of reasoning across comic sequences.
The central challenge can be represented as

$$P_{t+1} = f(P_1, P_2, \ldots, P_t), \tag{27}$$

where the interpretation of the next panel depends on information accumulated from earlier panels.
The more recent ComicsPAP benchmark by Vivoli et al. [10] extends this line of research. The

benchmark contains five tasks: sequence filling, text closure, visual closure, character coherence, and caption relevance. Its dataset contains 99,429 training samples, 1,226 validation samples, and 3,278 test samples.
Importantly, the study reports that contemporary large multimodal models can perform near chance on these comic-specific sequential understanding tasks in zero-shot settings. Fine-tuning substantially improves performance, demonstrating that general multimodal capability does not automatically imply effective comic-sequence reasoning.
This result is particularly significant for the present literature study because it suggests that simply applying a large vision-language model to a comic page may not be sufficient. The model must be capable of exploiting structural, contextual, and sequential relationships.

3.11 Vision-Language Models for Comics

Recent developments in large multimodal models have created new possibilities for comics understanding. Vision-language models can process image and textual inputs jointly and can potentially perform tasks ranging from visual question answering to narrative generation.
The survey by Vivoli et al. [1] places visual reasoning, visual question answering, image-to-text generation, and grounded captioning among higher-level comics-understanding tasks.
However, the ComicsPAP benchmark demonstrates an important limitation: strong performance on general-purpose multimodal tasks does not necessarily transfer to comic-specific reasoning [10]. Comics contain sequential dependencies, implicit events, unusual spatial relationships, and highly stylized visual representations. Consequently, comic-specific adaptation may be necessary even when using large pretrained multimodal models.
This observation motivates a hybrid research direction in which specialized structural modules are combined with multimodal models. Instead of expecting a single model to infer all page structure, character identities, dialogue associations, and narrative relations implicitly, the information can first be organized into explicit intermediate representations.

3.12 Narrative Generation

Narrative generation represents a higher-level objective in which information extracted from a sequence of comic panels is transformed into coherent natural-language prose.
Sachdeva et al. [8] proposed a system titled *From Panels to Prose*, which is particularly relevant to the research direction considered in this work. Their framework combines comic-element detection and association with transcript generation, panel captioning, character grounding, and prose generation.
The conceptual pipeline can be represented as

$$\text{Comic Page} \rightarrow \text{Elements} \rightarrow \text{Associations} \rightarrow \text{Panel Representation} \rightarrow \text{Narrative}. \tag{28}$$

An important aspect of this approach is that narrative generation is not treated as an independent image-captioning task. Instead, information about characters, speech, panels, and their relationships is incorporated before prose generation.
This distinction is important because a generated narrative should preserve character identity and event continuity. A system that generates independent captions for individual panels may describe visible content without correctly reconstructing the relationships between successive events.
Thus, the literature suggests that narrative generation should be positioned at the end of a pipeline that has already established sufficient structural and semantic grounding.

3.13 Ambiguity and Underspecification

An additional theoretical challenge is that multimodal narratives often contain information that is deliberately left unspecified.
Hutchinson et al. [11] examine underspecification in scene description-to-depiction tasks and emphasize that visual and textual representations do not encode all information in identical ways. Different modalities may leave different aspects of an event implicit.
This observation is particularly relevant to comics because the space between two panels, commonly called the gutter, may omit an action that the reader is expected to infer.
For example, if one panel shows a character holding an object and the next panel shows the object broken on the ground, the intermediate action may not be explicitly depicted. A computational system must therefore distinguish between information that is absent because of insufficient data and information that is intentionally omitted as part of the narrative structure.
The concept of underspecification consequently provides a useful theoretical perspective for understanding why comic narratives are difficult for automated systems. The goal is not necessarily to recover a single objectively correct missing event; rather, the system must construct an interpretation that is consistent with the available multimodal evidence.

3.14 Human Reading and Interface Considerations

Computational models can also benefit from understanding how humans interact with comic layouts. Chen et al. [5] investigated the effects of comic application interfaces on reader comprehension, cognitive processing, and narrative engagement.
Their study compared different presentation modes, including full-page multi-panel viewing, single-frame scrolling, and guided-view presentation. The findings suggest that the presentation of panels and preservation of page-level structure can affect narrative comprehension and engagement.
This work is relevant to computational research because it reinforces the importance of page-level organization. A system that processes each panel independently may lose information conveyed by the relationship between panels and by the overall page composition.
Therefore, page layout should be regarded not merely as a visual segmentation problem but as part of the narrative representation.

3.15 Global Variation in Comic Layouts

The work of Cohn et al. [4] further demonstrates that comic structure varies substantially across cultures and publication contexts.
The TINTIN corpus contains more than 76,000 panels from 1,030 comics representing 144 countries and territories. The analysis investigates panel frequency and size and reports systematic variation across styles, regions, language-related factors, and publication periods.
The observed relationship between panel frequency and panel size is especially relevant for computational analysis. Pages containing more panels tend to allocate less area to individual panels, while pages with fewer panels can contain substantially larger visual regions.
These findings imply that a model trained using a narrow collection of comic styles may encounter difficulties when applied to visually or culturally different comics. Generalization across styles and publication contexts is therefore an important consideration for future research.

3.16 Cartoon Image Processing as a Supporting Research Area

Although comics are a distinct medium, computational comics research also overlaps with the broader field of cartoon image processing.
Zhao et al. [7] provide a broad survey covering tasks such as quality assessment, style transfer, colorization, detection, recognition, image enhancement, emotion analysis, visualization, and interaction.
Cartoon images differ from natural photographs because they commonly contain strong contours, relatively uniform color regions, simplified textures, and stylized representations. These characteristics can affect the behavior of computer vision models originally developed for natural images.
This literature therefore provides supporting motivation for adapting vision techniques to stylized visual domains rather than assuming that methods developed exclusively on photographic datasets will transfer without modification.

3.17 Literature Synthesis

The reviewed literature reveals that computational comic understanding can be organized into a hierarchy of interdependent tasks.
At the first level, the system must understand the physical structure of the page. Panel segmentation methods such as that of Pang et al. [3] address this problem, while Cohn [6] shows that determining the reading order requires more than simple spatial sorting.
At the second level, the system must identify the semantic elements within individual panels, including characters, objects, and textual regions. Generic object detection methods and cartoon-processing research provide relevant visual foundations, while comics-specific research addresses character identity and text extraction.

At the third level, the visual and textual components must be associated. This includes identifying which speech balloon belongs to which character, grounding characters, and connecting dialogue with the visual scene.
At the fourth level, the system must perform multimodal reasoning. The literature on EmoComicNet and COMICS demonstrates that visual and textual information provide complementary evidence.
At the fifth level, the system must reason across multiple panels. Closure, character coherence, sequence completion, and contextual dependencies become central challenges. The ComicsPAP benchmark demonstrates that these capabilities remain difficult even for modern large multimodal models.
Finally, narrative generation can transform the structured information into a coherent textual description. The *From Panels to Prose* framework represents an important recent step toward this objective.
The resulting progression can be summarized as

$$\begin{aligned}\text{Page Structure} &\rightarrow \text{Panel Sequence} \rightarrow \text{Visual and Textual Elements} \rightarrow \text{Character–Text Association} \\ &\rightarrow \text{Multimodal Fusion} \rightarrow \text{Sequential Reasoning} \rightarrow \text{Narrative Reconstruction}.\end{aligned} \tag{29}$$

This synthesis forms the theoretical basis for the conceptual research framework presented in Section 4.

3.18 Literature Comparison Matrix

Table 1 summarizes the major contributions of the twelve papers studied in this study.

Table 1: Comparative summary of the literature reviewed

No.	Study	Primary Problem	Main Contribution	Research Layer
1	Iyyer et al. [2]	Narrative closure	COMICS dataset and multimodal cloze tasks	Sequential reasoning
2	Hutchinson et al. [11]	Multimodal underspecification	Conceptual analysis of ambiguity between modalities	Multimodal semantics
3	Vivoli et al. [1]	Comics understanding	LoCU taxonomy and survey of datasets and tasks	Overall field
4	Pang et al. [3]	Panel extraction	Adaptive recursive panel segmentation	Page layout
5	Cohn [6]	Reading order	Empirical and theoretical model of page navigation	Layout and sequence

No.	Study	Primary Problem	Main Contribution	Research Layer
6	Dutta et al. [12]	Emotion recognition	Multimodal relevance-aware emotion model	Semantic understanding
7	Zhao et al. [7]	Cartoon processing	Survey of computer vision methods for cartoon images	Vision foundation
8	Chen et al. [5]	Comic reading interfaces	Study of layout, comprehension, and engagement	Human cognition
9	Cohn et al. [4]	Global panel structure	Large-scale cross-cultural analysis of comic panels	Layout and generalization
10	Gupta et al. [9]	Object detection	Optimized YOLO-NAS-based detection framework	Visual detection
11	Sachdeva et al. [8]	Comic-to-prose generation	Integrated comic understanding and narrative generation	Narrative generation
12	Vivoli et al. [10]	Sequential comic understanding	Five-task benchmark for comic-strip reasoning	Multimodal reasoning

4 Research Gap and Conceptual Framework

4.1 Introduction

The literature reviewed in the previous section demonstrates that computational understanding of comics is a highly interdisciplinary problem involving computer vision, optical character recognition, natural language processing, multimodal learning, and sequential reasoning. Existing studies have addressed several individual components of comic understanding, including panel extraction, reading-order determination, character and object recognition, text extraction, emotion recognition, multimodal reasoning, and narrative generation [3, 6, 12, 2, 8].

However, the reviewed literature also indicates that understanding a comic as a complete visual narrative remains substantially more difficult than solving any single component independently. A comic page contains multiple interacting information sources: panels, characters, objects, visual actions, speech bubbles, captions, and other textual or graphical elements. These elements must be interpreted not only individually but also in relation to one another and across successive panels.

The present section synthesizes the findings from the literature and develops a conceptual research framework for multimodal understanding of comics and visual narratives. The framework is derived

from the capabilities and limitations identified in the reviewed studies. It is therefore important to emphasize that the framework represents a proposed research direction rather than an implemented system.

4.2 Research Gap

The literature review identifies several important research gaps in the current state of computational comic understanding.

4.2.1 Fragmentation of Comic Understanding Tasks

A major observation from the literature is that comic understanding has traditionally been divided into several specialized tasks. These include panel segmentation, reading-order prediction, object and character detection, OCR, character–text association, emotion recognition, and narrative understanding.

For example, Pang et al. focused primarily on extracting individual panels from manga pages [3], while Cohn investigated the cognitive and structural principles underlying navigation through comic page layouts [6]. Similarly, EmoComicNet concentrated on multimodal emotion recognition [12]. These studies provide important building blocks, but they do not by themselves constitute a complete system for interpreting the narrative represented by a comic page.

Recent research has attempted to combine several of these capabilities. In particular, the work presented in *From Panels to Prose* integrates panel and character-related processing, text extraction, character grounding, and narrative generation [8]. Nevertheless, the broader literature indicates that comprehensive comic understanding remains an open research problem.

4.2.2 Difficulty of Determining Reading Order

Reading order is another important challenge. A comic page is not necessarily organized according to a simple left-to-right or top-to-bottom sequence. Irregular panel layouts, overlapping panels, unconventional compositions, and culturally different reading conventions can influence how a page is interpreted.

Cohn's empirical investigation demonstrated that readers employ structural strategies for navigating comic pages rather than simply applying a universal spatial sorting rule [6]. The large-scale TINTIN corpus study further demonstrates substantial variation in panel organization across different comics, regions, languages, and publication contexts [4].

Consequently, reading order should be regarded as a structural reasoning problem rather than merely a geometric sorting problem.

4.2.3 Insufficiency of Individual Modalities

Another recurring finding is that visual and textual information cannot always be interpreted independently. Images provide information about characters, objects, actions, environments, and visual expressions, whereas text may describe dialogue, thoughts, narration, or contextual information.
The COMICS work by Iyyer et al. demonstrated that comic-story understanding requires reasoning across both visual and textual information [2]. Similarly, multimodal emotion recognition research has shown that combining visual and textual information can provide advantages over using either modality independently [12].
This suggests that a complete comic-understanding system requires explicit multimodal integration rather than treating OCR and visual analysis as independent terminal tasks.

4.2.4 Cross-Panel Reasoning and Narrative Closure

A particularly significant research gap concerns the interpretation of information that is distributed across multiple panels. Comic narratives often do not explicitly show every event. Instead, readers infer transitions, actions, consequences, and relationships from the sequence of panels.
This phenomenon is central to the concept of the "gutter," where information between two panels may be visually absent but narratively meaningful. Iyyer et al. demonstrated this problem through multimodal cloze and character coherence tasks designed to test reasoning across panels [2].
The problem is also related to the broader concept of underspecification in multimodal systems. Textual and visual representations can leave different aspects of an event unspecified, meaning that an interpretation may require reasoning beyond the information explicitly present in either modality [11].
Therefore, narrative understanding requires more than recognizing the content of individual panels. It requires maintaining contextual information across the sequence.

4.2.5 Challenges for Large Vision–Language Models

Recent multimodal and vision–language models have significantly expanded the possibilities for visual reasoning. However, the ComicsPAP benchmark indicates that general-purpose large multimodal models can still struggle with comic specific tasks involving sequential and contextual reasoning [10].
The literature therefore suggests that increasing model size alone may not solve the problem. Comic understanding involves domain-specific structures, including panel relationships, character recurrence, speech attribution, reading order, and implicit narrative transitions.
The recent survey on comics understanding similarly organizes the field into multiple layers and task groups, demonstrating the breadth of capabilities required for comprehensive comic interpretation [1].

4.2.6 Dataset and Annotation Challenges

The reviewed literature also highlights limitations related to datasets and annotations. Comic datasets differ substantially in artistic style, language, layout, publication period, and cultural conventions. Furthermore, different datasets provide different levels of annotation, such as panel boundaries, characters, dialogue, emotions, or narrative relationships.
The survey literature identifies dataset accessibility and the lack of standardized annotations as important challenges for comics understanding [1]. The global analysis of comic panels further demonstrates the considerable structural diversity present across comics [4].
This heterogeneity creates difficulties for developing models that generalize across different comic styles and cultural contexts.

4.2.7 Need for Grounded Narrative Generation

The final research gap concerns the transition from low-level recognition to high-level narrative interpretation. Detecting a character or extracting a piece of dialogue does not automatically establish what happened in the story.
The *From Panels to Prose* study represents an important step toward this direction by combining visual and textual processing with character grounding and prose generation [8]. However, coherent narrative generation remains dependent on the quality of preceding structural and multimodal reasoning stages.
A future system should therefore generate narrative descriptions that remain grounded in detected characters, textual information, visual events, and the relationships between consecutive panels.

4.3 Problem Decomposition

Based on the literature, comic understanding can be decomposed into a sequence of interconnected problems:

1. **Page and panel analysis:** identifying the structural organization of a comic page.
2. **Reading-order determination:** establishing the sequence in which panels should be interpreted.
3. **Visual understanding:** identifying characters, objects, scenes, actions, and visual expressions.
4. **Text understanding:** detecting textual regions and extracting their contents using OCR and related techniques.
5. **Character–text association:** determining which character corresponds to a particular speech bubble or textual element.

6. **Multimodal fusion:** integrating information obtained from visual and textual modalities.
7. **Sequential reasoning:** relating information across consecutive panels and resolving implicit transitions.
8. **Narrative reconstruction:** converting the integrated representation into a coherent description of the visual narrative.

This decomposition does not imply that the components must necessarily be implemented as independent modules. Rather, it provides a conceptual organization of the capabilities identified across the reviewed literature.

4.4 Conceptual Research Framework

Based on the above research gaps, a conceptual framework can be formulated for multimodal comic understanding. The framework begins with a comic page as input and progressively transforms its visual and textual information into a structured narrative representation.
The conceptual framework is shown below.

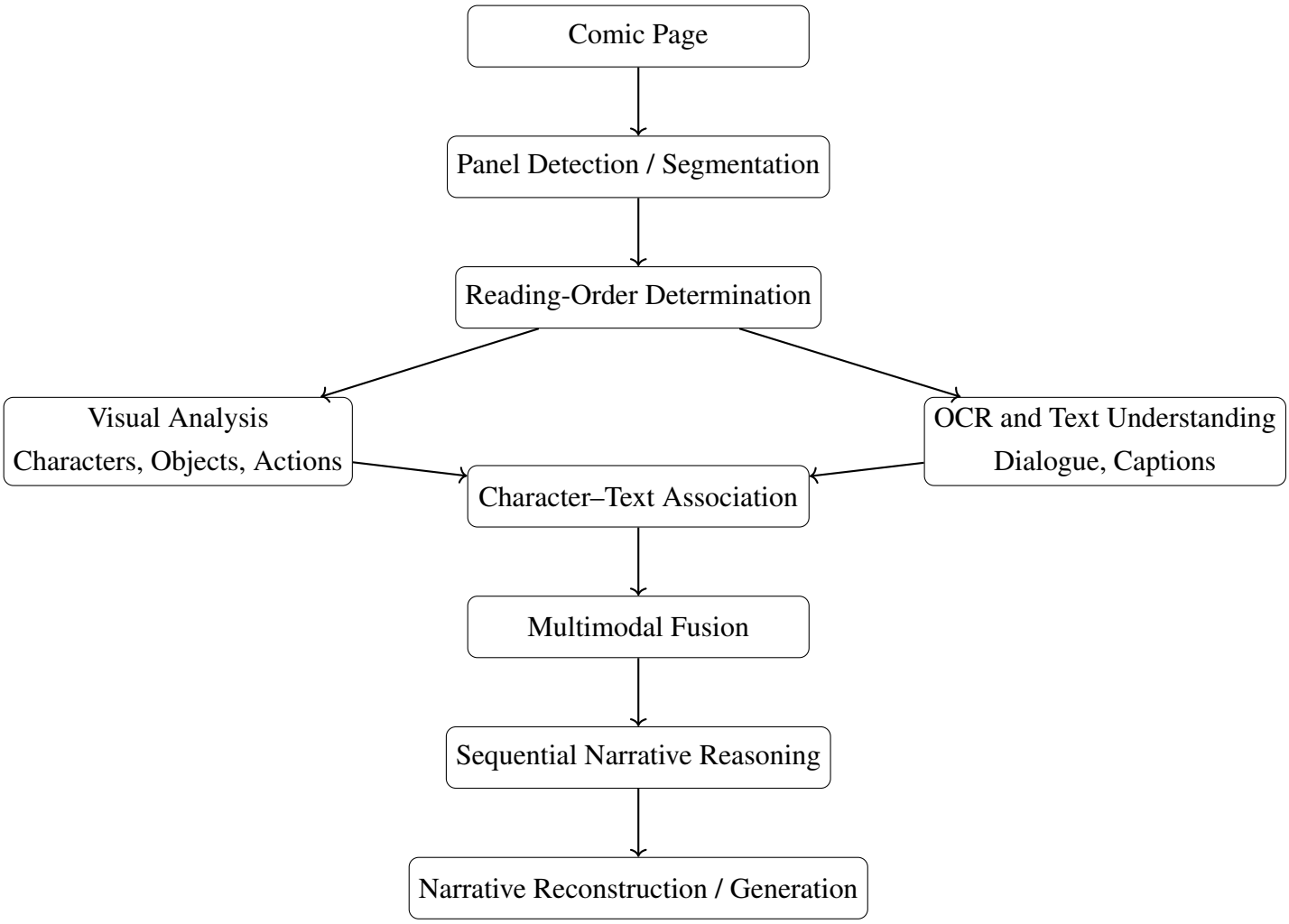


Figure 2: Conceptual Research Framework Derived from the Literature

The framework should be interpreted as a research formulation derived from the reviewed studies. It was not implemented or experimentally evaluated during the present work.

4.5 Multimodal Information Fusion

The multimodal fusion stage represents a central component of the proposed conceptual framework. Information extracted from comic images and information obtained from textual elements provide complementary descriptions of the same narrative.
For example, a visual region may indicate that a character is present and performing an action, while an associated speech bubble may provide information about the character's intention or response. Combining these signals can provide a more complete representation than either modality alone.
A conceptual multimodal representation may therefore be expressed as

$$\mathbf{z}_i = F_v(\mathbf{v}_i) \oplus F_t(\mathbf{t}_i), \tag{30}$$

where $\mathbf{v}_i$ represents visual information from panel i, $\mathbf{t}_i$ represents textual information, F_v and F_t denote modality-specific representations, and $\oplus$ represents a conceptual fusion operation.
The equation is intended only as a conceptual representation of multimodal integration. It does not specify an implemented architecture.
The importance of such integration is supported by the COMICS study and by multimodal emotion-recognition research, both of which demonstrate that visual and textual information can contribute complementary evidence [2, 12].

4.6 Sequential Narrative Reasoning

After multimodal information has been extracted and associated within individual panels, the next requirement is to establish relationships between panels.
Let a comic contain an ordered sequence

$$P = \{p_1, p_2, \ldots, p_n\}. \tag{31}$$

Each panel can be represented conceptually as

$$p_i = (V_i, T_i, C_i, A_i), \tag{32}$$

where:

- V_i represents visual information,
- T_i represents textual information,
- C_i represents character-related information, and

- A_i represents actions or events.

The narrative interpretation of panel p_i should not necessarily depend only on the information contained in p_i. Instead, its interpretation can depend on preceding and succeeding contextual information:

$$N_i = f(p_i, p_{i-1}, p_{i+1}, H_i), \tag{33}$$

where H_i represents the contextual history available at panel i.
This formulation captures an important characteristic of comics: meaning may be distributed across multiple panels. In particular, the gutter between panels can contain implicit narrative information that must be inferred from context [2].

4.7 Narrative Reconstruction

The final stage of the conceptual framework is narrative reconstruction. Instead of treating each panel as an isolated image, the system envisioned by the framework would transform the accumulated multimodal representation into a coherent description.
A conceptual representation is

$$R = G(Z_1, Z_2, \ldots, Z_n), \tag{34}$$

where Z_i represents the integrated multimodal representation of panel i and G represents a future narrative reconstruction or generation mechanism.
The generated narrative should ideally preserve:

1. the identity and recurrence of characters;
2. the relationship between characters and dialogue;
3. the sequence of events;
4. important objects and visual actions;
5. emotional or contextual information when supported by the input; and
6. implicit relationships between successive panels.

The recent work on generating prose narratives from comics demonstrates the feasibility and relevance of this overall direction, while also illustrating the number of intermediate capabilities required before reliable narrative generation can be achieved [8].

4.8 Relationship Between the Reviewed Literature and the Framework

The proposed conceptual framework synthesizes different contributions from the reviewed literature.

Table 2: Mapping of Literature Contributions to the Conceptual Framework

Framework Component	**Representative Literature**	**Contribution**
Panel segmentation	Pang et al. [3]	Automatic extraction and structural analysis of comic panels
Reading order	Cohn [6]	Empirical analysis of how readers navigate comic page layouts
Global layout variation	Cohn et al. [4]	Evidence of structural diversity across comics from different regions and contexts
Visual processing	Zhao et al. [7]	Broader computer-vision foundations for stylized and cartoon imagery
Object detection	Gupta et al. [9]	General object-detection methodology relevant to potential visual-analysis stages
Multimodal understanding	Iyyer et al. [2]	Joint visual-textual reasoning for comic narrative understanding
Emotion and context	Dutta et al. [12]	Multimodal integration of visual and textual information for emotion recognition
Sequential reasoning	Iyyer et al. [2]	Cross-panel inference and narrative closure
Multimodal benchmark	Vivoli et al. [10]	Evaluation of comic-specific multimodal and sequential reasoning capabilities
Narrative generation	Sachdeva et al. [8]	Integration of comic understanding components toward prose generation
Ambiguity	Hutchinson et al. [11]	Conceptual basis for understanding underspecified multimodal information
Field taxonomy	Vivoli et al. [1]	Comprehensive organization of comic-understanding tasks and research layers

The mapping demonstrates that the framework is not based on a single existing method. Instead, it synthesizes complementary findings from multiple areas of research.

4.9 Scope of the Present Work

The present work focuses on the literature-based analysis of computational approaches for multimodal understanding of comics and visual narratives. The study examines the major components required for computational comic understanding, including page and panel analysis, reading-order determination, visual and textual understanding, character–text association, multimodal fusion, sequential reasoning, and narrative reconstruction.
Based on the synthesis of these research directions, a conceptual framework is formulated to organize the identified capabilities into an integrated research pipeline. The framework is intended as a research formulation that can guide subsequent implementation and experimental investigation. Experimental implementation and quantitative evaluation are outside the scope of the present study. Consequently, the framework should be interpreted as a literature-derived conceptual research framework rather than an experimentally validated architecture.

5 Discussion and Literature-Based Findings

5.1 Introduction

The literature reviewed in this research work demonstrates that computational understanding of comics is not a single computer-vision or natural-language processing task. Instead, it involves a combination of structural, visual, textual, multimodal, and sequential reasoning capabilities.
The twelve studies considered in this work collectively cover a broad range of these capabilities. They include panel segmentation, reading-order analysis, character and object understanding, optical character recognition, multimodal emotion recognition, narrative closure, multimodal benchmarking, narrative generation, and broader research on cartoon-image processing [3, 6, 12, 2, 8, 1].
This section discusses the principal findings that emerge from the literature and examines how the different research directions contribute to the proposed conceptual framework. Since the present work was limited to literature study and research formulation, the findings discussed in this section are literature-based findings and do not represent experimentally obtained results from the present work.

5.2 Major Findings from the Literature

5.2.1 Comic Understanding is Fundamentally Multimodal

One of the recurring findings from the reviewed literature is that comics cannot be adequately represented through visual information alone. A comic page simultaneously contains images, text, spatial layout, character information, and relationships between successive panels.
The COMICS study demonstrated this issue through multimodal narrative understanding tasks, where both textual and visual information contribute to understanding the story [2]. Similarly, EmoComicNet demonstrated the usefulness of combining visual and textual information for emotion recognition in comics [12].
Therefore, a computational system intended to understand comics should not treat OCR, visual recognition, and narrative interpretation as completely independent problems. Their outputs should ultimately contribute to a shared multimodal representation.
This observation directly motivates the multimodal-fusion stage of the conceptual framework proposed in Section 4.

5.2.2 Panel Structure Provides the Foundation for Higher-Level Understanding

The reviewed literature indicates that panel structure is one of the fundamental organizational properties of a comic page.
Pang et al. investigated automatic panel extraction and demonstrated the difficulty of dealing with irregular, unclosed, borderless, and joined panels [3]. These observations show that the visual segmentation of a comic page is itself a non-trivial problem.
The global analysis presented by Cohn et al. further indicates that panel organization varies substantially across comics from different geographical, linguistic, and publication contexts [4]. Consequently, panel segmentation should not simply be considered a preprocessing operation. Errors in identifying panels can propagate into subsequent stages, including reading-order determination, OCR, character tracking, and narrative reasoning.

5.2.3 Reading Order Cannot Always be Determined by Simple Spatial Rules

Another important finding is that comic reading order is influenced by the structural organization of a page.
Cohn's empirical study showed that readers do not simply follow a universal geometric rule such as always scanning from left to right and then from top to bottom [6]. Instead, readers use structural constraints and preferences to navigate complex layouts.
The TINTIN corpus study also provides evidence for substantial variation in panel size and frequency across different comic traditions [4].
This has an important implication for computational systems: reading-order prediction should incorporate page structure and contextual relationships rather than relying exclusively on panel

coordinates.

5.2.4 Character Understanding is Central to Narrative Interpretation

Characters provide continuity across a comic sequence. A character detected in one panel may reappear in several later panels, potentially in different poses, locations, or visual contexts.
The literature on comic understanding therefore treats character recurrence, character coherence, and character grounding as important components of narrative interpretation [2, 1].
The work of Sachdeva et al. is particularly relevant because it combines character detection, association, clustering, and grounding as part of a larger comic-to-prose pipeline [8].
This suggests that character recognition should not be viewed merely as an object-detection problem. For narrative understanding, the system must also maintain the identity and contextual role of characters across panels.

5.2.5 Text Extraction Alone is Insufficient

Textual elements in comics contain dialogue, narration, captions, and other information that can be essential for interpreting a scene. However, simply extracting the text does not necessarily determine its meaning.
A speech bubble must be associated with the appropriate speaker, and the speaker's identity must be understood in relation to the visual scene.
The literature therefore suggests that OCR should form one component of a larger text-understanding pipeline rather than being treated as the final objective [1, 8].
The comic-to-prose framework proposed by Sachdeva et al. illustrates this principle by incorporating text extraction and character–dialogue association before narrative generation [8].

5.2.6 Cross-Panel Reasoning is a Major Remaining Challenge

The concept of narrative closure is particularly important for comics because events are frequently distributed across multiple panels.
The gutter between two panels can omit an event while still allowing a human reader to infer what happened. Iyyer et al. explicitly investigated this problem through tasks requiring inference across panels [2].
The finding is significant because it separates comic understanding from ordinary image captioning. Describing each panel independently does not necessarily reconstruct the story represented by the complete page.
A successful future system must therefore maintain contextual information across the panel sequence and reason about relationships between events.

5.2.7 Large Vision–Language Models Do Not Automatically Solve Comic-Specific Reasoning

Recent vision–language models provide increasingly powerful capabilities for image and text understanding. However, the ComicsPAP benchmark indicates that general-purpose large multimodal models can still face difficulties on comic-specific tasks involving sequential and contextual understanding [10].
An important observation from this work is that model scale alone does not guarantee reliable comic understanding. The benchmark evaluates tasks such as sequence filling, text closure, visual closure, character coherence, and caption relevance, which require domain-specific reasoning.
This supports the broader conclusion of the comics-understanding survey that the field involves multiple specialized layers and task categories rather than one general visual question-answering problem [1].

5.2.8 Dataset Diversity is Both an Opportunity and a Challenge

The literature demonstrates considerable diversity in comic datasets. Comics differ in artistic style, panel organization, language, cultural conventions, publication period, and level of annotation.
The TINTIN corpus provides large-scale evidence of global variation in comic panel structures [4].
The comics-understanding survey also identifies dataset accessibility and annotation availability as important issues for future research [1].
This suggests that models developed using a single narrow dataset may not necessarily generalize to other comic styles or cultural traditions.
Future experimental research should therefore consider dataset diversity and cross-domain generalization as important evaluation criteria.

5.3 Comparison of Existing Approaches

The reviewed papers can be broadly categorized according to the level of comic understanding they address.

Table 3: Comparison of Major Research Directions in the Reviewed Literature

Research Direction	**Representative Work**	**Primary Strength**	**Main Limitation**
Panel extraction	Pang et al. [3]	Handles difficult comic panel boundaries	Focused primarily on structural extraction
Reading-order analysis	Cohn [6]	Provides empirical understanding of page navigation	Human reading principles do not directly constitute an implemented AI system

Research Direction	Representative Work	Primary Strength	Main Limitation
Global comic analysis	Cohn et al. [4]	Demonstrates large-scale structural diversity	Describes corpus-level properties rather than complete narrative understanding
Multimodal narrative reasoning	Iyyer et al. [2]	Tests inference across visual and textual context	Focused on specific benchmark tasks
Emotion recognition	Dutta et al. [12]	Combines visual and textual modalities	Targets emotion rather than complete narrative reconstruction
Comic benchmark	Vivoli et al. [10]	Evaluates modern multimodal models on comic-specific tasks	Benchmark performance remains challenging even for large models
Survey and taxonomy	Vivoli et al. [1]	Provides broad organization of comics-understanding research	Does not itself provide a complete end-to-end implementation
Narrative generation	Sachdeva et al. [8]	Connects structural understanding with prose generation	Narrative generation depends on the reliability of multiple upstream stages

The comparison shows that different studies solve different portions of the overall problem. The research direction proposed in this report therefore builds upon their complementary contributions rather than attempting to replace any individual approach.

5.4 Challenges Identified from the Literature

5.4.1 Irregular Visual Layouts

Comic pages can contain panels with different sizes and shapes, irregular boundaries, overlapping regions, and non-standard spatial arrangements [3, 6].
Such layouts make conventional object-detection and document-analysis assumptions less reliable.

5.4.2 Stylized Visual Content

Cartoon and comic images differ substantially from natural photographs. They often contain line drawings, flat regions of color, exaggerated facial features, simplified objects, and artistic distortions.

The survey by Zhao et al. emphasizes the distinctive properties of cartoon images and the resulting challenges for conventional computer-vision approaches [7].
Therefore, models designed primarily for natural-image datasets may require adaptation when applied to comic imagery.

5.4.3 OCR in Comic Images

Text appearing in comics can vary considerably in font, size, orientation, placement, and visual style. It may also be embedded in complex backgrounds.
The literature therefore treats text extraction as a challenging component of comic understanding rather than a trivial preprocessing step [2, 1].

5.4.4 Character–Dialogue Association

A dialogue bubble does not necessarily provide an explicit textual indication of its speaker.
Correctly associating text with a character may require understanding spatial relationships, speech-bubble tails, character positions, and the surrounding visual context.
This problem is particularly relevant to narrative generation because incorrect speaker attribution can alter the meaning of an entire scene [8].

5.4.5 Implicit Events and Narrative Gaps

Comics frequently communicate events indirectly. The transition from one panel to another may imply an action that is never explicitly depicted.
This makes narrative reconstruction fundamentally different from independently captioning images.
A future system must infer relationships while avoiding unsupported assumptions.
The issue is closely related to the concept of underspecification discussed in the multimodal literature [11].

5.4.6 Cultural and Linguistic Variation

Comic reading conventions are not globally uniform. The direction and structure of reading may depend on cultural and linguistic conventions [6].
The large-scale TINTIN analysis reinforces the importance of considering global variation rather than assuming that one comic tradition represents all comics [4].

5.4.7 Evaluation of Complete Narrative Understanding

Evaluating a complete narrative understanding system is more difficult than evaluating an isolated detector.
Metrics such as detection accuracy or OCR accuracy provide useful information about individual components, but they do not necessarily indicate whether the system correctly understood the

overall story.
The emergence of comic-specific benchmarks such as ComicsPAP reflects the need for evaluations that directly test sequential and multimodal understanding [10].

5.5 Implications for Future Research

The findings from the reviewed literature suggest several important directions for future experimental research. An integrated multimodal pipeline should investigate the joint use of panel analysis, visual understanding, OCR, character tracking, and textual interpretation rather than treating these components as completely independent tasks. Reading-order prediction should combine spatial structure with contextual and semantic information, while persistent character representations could support character continuity across panels. Future systems should also investigate sequential multimodal reasoning to distinguish explicitly visible information, textual information, inherited context, and events that are implicitly inferred between panels. In addition, evaluation should extend beyond individual component-level measures to include comic-specific capabilities such as visual and textual closure, character coherence, sequential reasoning, narrative consistency, and the quality of generated descriptions. These directions indicate that future progress in comic understanding will require models that integrate visual, textual, structural, and sequential information within a unified reasoning framework.

6 Limitations, Future Work and Conclusion

6.1 Limitations of the Present Study

The present work is a literature-based study and therefore has limitations associated with its research scope. The analysis is based on the selected literature and does not include an independently conducted experimental evaluation.
In particular, the conceptual framework developed in this work has not been implemented as an executable computational system. Consequently, no model training, dataset-specific testing, quantitative benchmarking, or empirical comparison with existing approaches is presented.
These limitations do not affect the purpose of the study, which is to synthesize existing research, identify important research gaps, and formulate a structured framework for future experimental investigation.

6.2 Proposed Future Experimental Methodology

The literature review provides a foundation for future experimental investigation of the conceptual framework developed in Section 4. A possible experimental pipeline can be organized as

$$\begin{aligned}\text{Dataset Preparation} &\rightarrow \text{Panel Analysis} \rightarrow \text{Reading Order} \rightarrow \text{Visual and Textual Analysis} \\ &\rightarrow \text{Multimodal Fusion} \rightarrow \text{Sequential Reasoning} \rightarrow \text{Narrative Generation}.\end{aligned} \tag{35}$$

The following stages describe a possible implementation strategy.

6.2.1 Dataset Acquisition and Preparation

An appropriate comic dataset would first need to be obtained for experimental evaluation. Depending on the research objectives, the dataset could contain annotations for panel boundaries, reading order, characters, dialogue, character–text associations, and narrative relationships.
The selection of the dataset should consider diversity in artistic style, language, layout, and cultural context. This is important because the literature demonstrates substantial variation across comic traditions.

6.2.2 Panel Detection and Segmentation

The first computational stage would identify individual panels from each comic page. Panel segmentation could investigate approaches capable of handling rectangular, irregular, borderless, overlapping, and joined panel structures.
The output of this stage would consist of panel regions while preserving their original spatial relationships within the page. Accurate panel extraction is important because errors at this stage may propagate to subsequent OCR, character analysis, reading-order estimation, and narrative reasoning.

6.2.3 Reading-Order Determination

Following panel extraction, the system would determine the sequence in which the panels should be interpreted. Rather than relying exclusively on left-to-right and top-to-bottom spatial sorting, future approaches could combine panel geometry, spatial relationships, structural grouping, and contextual information.
The system should also be evaluated across different comic traditions to determine whether the learned ordering principles generalize beyond a particular dataset or reading convention.

6.2.4 Visual Analysis

The visual-analysis stage would identify the important visual elements within each panel. Potential components include character detection, object detection, scene understanding, action recognition, expression analysis, and character recurrence across panels.
Generic object-detection methods may provide a starting point for this stage. However, the stylized and highly variable nature of comic imagery makes comic-specific adaptation an important area for future investigation.

6.2.5 OCR and Text Understanding

The textual-analysis stage would identify and extract dialogue, captions, narration, signs, inscriptions, and other textual elements. OCR should be combined with text-region detection and spatial information so that extracted text can be associated with its corresponding location and surrounding visual context.
The resulting textual representation would subsequently be integrated with visual and character information rather than being treated as an independent output.

6.2.6 Character–Text Association

The system would next establish relationships between characters and textual elements, particularly speech balloons and dialogue. This stage would require determining which character is most likely to be associated with a given textual element.
Character identity should also be maintained across panels so that recurring characters can be represented consistently throughout the narrative.

6.2.7 Multimodal Fusion

Visual, textual, spatial, and character-level information would then be combined into an integrated multimodal representation.
The objective would not simply be to concatenate different feature representations, but to allow information from one modality to assist in interpreting another. For example, visual information could identify a character and its action, while dialogue could provide information about the character's intention or response.

6.2.8 Sequential Narrative Reasoning

The integrated representations would subsequently be processed across the ordered sequence of panels. This stage would focus on identifying relationships between events, maintaining character continuity, and reasoning about information that is distributed across multiple panels.
Particular attention should be given to implicit events occurring between panels. A future system could investigate memory mechanisms, temporal representations, or multimodal sequential architectures for improving narrative closure and contextual reasoning.

6.2.9 Narrative Generation

The final stage would transform the integrated multimodal representation into a coherent narrative description.
The generated narrative should preserve character identity, dialogue relationships, event order, important visual actions, contextual information, and relationships between successive panels.

Narrative generation should also remain grounded in the visual and textual evidence available in the source comic to reduce unsupported or hallucinated information.

6.3 Potential Evaluation Methodology

A future implementation of the proposed framework would require evaluation at both individual task levels and the overall narrative-understanding level.
Panel detection and segmentation could be evaluated using measures such as intersection-over-union and related segmentation metrics. Reading-order prediction could be evaluated by comparing the predicted sequence with annotated panel order. Visual and character understanding could be assessed using appropriate detection, recognition, and association metrics.
OCR performance could be evaluated using character- or word-level recognition measures.
Multimodal reasoning could be evaluated using comic-specific tasks such as sequence completion, visual closure, text closure, character coherence, and contextual relevance.
Narrative generation would require a combination of automatic and human evaluation. Important criteria could include factual consistency with the source comic, character consistency, event-order consistency, visual and textual grounding, narrative coherence, and completeness. Human evaluation would be particularly useful because conventional text-generation metrics may not fully capture whether a generated narrative faithfully represents the original visual story.

6.4 Future Research Directions

The literature reviewed in this work suggests several promising directions for further research.
Future work could investigate the development of comic-specific vision–language models through domain-specific fine-tuning, instruction tuning, or specialized multimodal representations. The difficulty of current multimodal models on comic-specific sequential tasks also suggests the need for models capable of better cross-panel contextual reasoning.
Another important direction is cross-cultural comic understanding. Since comic styles, layouts, languages, and reading conventions vary substantially, future systems should be evaluated across diverse comic traditions rather than being restricted to a single dataset or visual style. Such evaluation could help determine whether a model learns general principles of visual narrative understanding or primarily learns dataset-specific conventions.
Improved narrative closure is also an important research direction. Future models could investigate explicit temporal representations, memory mechanisms, and multimodal sequential architectures to improve the inference of events that are implicitly represented between panels.
Future narrative-generation systems should additionally prioritize grounding over purely fluent language generation. Generated descriptions should remain supported by identifiable visual and textual evidence from the source comic, thereby reducing unsupported characters, events, relationships, and actions.
Human–AI interaction provides another potential direction in which multimodal systems could

assist users in navigating, querying, and explaining complex comic narratives. Such systems could combine visual question answering, character tracking, contextual retrieval, and narrative explanation.

Finally, the principles developed for comic understanding may potentially be extended to other forms of multimodal visual storytelling, including museum collections and historical visual archives. In such applications, images could be interpreted jointly with captions, inscriptions, metadata, and contextual information to construct richer cultural or historical narratives. This extension would require domain-specific datasets, annotations, and evaluation methodologies and therefore represents a longer-term research possibility.

6.5 Conclusion

The present work examined the field of multimodal understanding of comics and visual narratives through a study of twelve relevant research papers. The literature demonstrates that comic understanding involves several interdependent tasks, including panel analysis, reading-order determination, visual and textual understanding, character–text association, multimodal fusion, sequential reasoning, and narrative reconstruction.

The review also identified persistent challenges such as irregular layouts, stylized visual content, OCR difficulties, character continuity, implicit information between panels, dataset heterogeneity, and the limitations of current multimodal models in sequential narrative reasoning.

Based on these findings, a conceptual research framework was formulated to organize these capabilities into an integrated pipeline. The framework provides a structured research direction connecting low-level visual and textual analysis with higher-level multimodal and sequential narrative reasoning.

The present study is literature-based and does not include experimental implementation or quantitative evaluation. Nevertheless, the systematic synthesis of existing research, identification of research gaps, and formulation of the conceptual framework provide a foundation for future experimental investigation in multimodal comic and visual-narrative understanding.

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